SEMINAR III · WEEK 5 · FOUNDATIONS

Percent

Fraction ↔ Decimal ↔ Percent, e.g. 1/2 = 0.5 = 50%

Two teaching sessions this week, not four separate lessons.

SESSION A

What Percent Means, and Finding Part / Percent / Whole

PERCENT MEANS OUT OF 100

25% = 25/100 | = 0.25 | = 1/4

The representation changes across fraction/decimal/percent — the quantity doesn't.

COMMON MISCONCEPTION

Is 20% just "20"?

> Wrong: 20% = 20. Right: 20% = 0.20 (or 20/100).

COMMON MISCONCEPTION

Which way does the decimal move?

> 35% → 3.5 is wrong. 35% < 100%, so its decimal form must be less than 1: 0.35.

Reasonableness catches this one every time — use it.

THREE STRUCTURES

Part, Percent, or Whole?

Find	How
the Part	percent × whole
the Percent	part ÷ whole, then convert
the Whole	division or benchmark reasoning

Before calculating, students should name what's known and what's missing.

COMMON MISCONCEPTION

18 of 24 students finished. What percent?

> Wrong: 24 ÷ 18. Right: 18 ÷ 24 — part (18) over whole (24).

EXIT TICKET — SESSION A

Part, percent, or whole?
Identify it — then solve.

SESSION B

Percent in Money, and Percent Change

MAJOR COMPREHENSION FOCUS

Label what you're finding before you calculate.

Original Price | Discount | Sale Price

Original Cost | Tax | Final Cost

Activity: "What Will I Actually Pay?" — original price → percent → estimate → calculate → check reasonableness.

COMMON MISCONCEPTION

Does a discount get added?

> $80 item, $16 discount. Wrong: 80+16=96. Right: 80−16=**$64** — a discount reduces the price.

COMMON MISCONCEPTION

Does tax get subtracted?

> \$50 purchase, \$5 tax. Wrong: $50-5=45$. Right: $50+5=$**\$55** — tax increases what's owed.

TIP PRACTICE

Fast mental strategies

10% 15% 20% 25%

Not every percent needs the full formula — 50% of $40 is just half.
Efficiency matters, not just correctness.

PERCENT CHANGE

original → new → difference → percent change

Absolute change (the difference) is not the same thing as percent change.

COMMON MISCONCEPTION

A value rises from 40 to 50. What's the percent change?

> Wrong: 10÷50. Right: 10÷40 — the **original** value is always the denominator.

COMMON MISCONCEPTION

A 20% increase, then a 20% decrease — back to the start?

> *No. The reference whole changes after the first calculation.*

CLOSING THIS WEEK

- Mixed practice: conversions, part/percent/whole, discounts, tax, tips, percent change
- Error analysis + retry, then the Week 5 check
- Reflection: one percent skill you can use outside school, one still developing, one error, one checking strategy

TRANSITION TO WEEK 6

Ratios, Rates & Proportions

Continue retrieving fractions, decimals, percent, division, estimation, and reasonableness. The big Week 6 question: what two quantities am I comparing?